Beginner challenges and our solution

AI Tennis Coach: personalized guidance between lessons and a plan for the next practice

Beginner difficulty	Our proposed response
Expensive coaching S$100+/h [2]. Two hours/week costs S$800+ over 4 weeks, about 27% of a S$3,000 stipend [1].	**Support between lessons** Review practice clips and give a clear correction before the next coach visit.
Unclear mistakes After a missed forehand: is it timing, balance or swing? Generic advice may not fit my movement.	**Personalized, explainable feedback** Use pose estimates and coaching references to show what to fix and why.
Limited practice opportunities NTU personal quota: one booking/7 days, max. 1 hour [2]. A slot must fit my partner and study schedule.	**Adaptable training plan** Plan around court access, time and partners, with solo drills between court sessions.

Budget example: international PhD student on the 2026 NTU Research Scholarship [1]

[1] NTU Research Scholarship

[2] NTU SRC booking rules

How our four stages work together

Each subgroup produces the input needed by the next stage

1. Data Collection

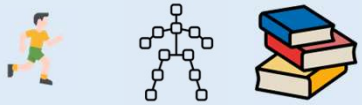


Collect a tennis clip, player level and goal. Record time and court access.

Output

Video + player context

2. Data Processing



Identify the stroke and pose.
Retrieve relevant coaching references.

Output

Movement features + reference material

3. AI Reasoning



Compare movement with references. Explain and prioritise the issue.

Output

Key issue + explanation

4. Output to User



Show feedback linked to the clip and recommend an adaptive practice plan.

Output

Feedback report + next-session plan

Prototype: forehand first, using 9 supplied tennis clips, 15 example prompts and coaching references.

Data Collection

User Input → **Data Processing** → **AI Reasoning** → **Output to User**

Tennis clips and player questions for a first prototype

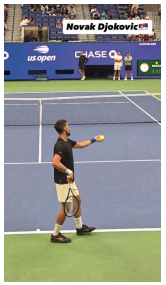
Motion examples



Forehand



Backhand



Serve

9 MP4 clips (3 per stroke)
2 GIFs and 2 JPGs

User context

15 example prompts

Example: beginner forehand

Goal: improve consistency

“Why is my forehand unstable and what should I focus on first?”

References: tennis.docx and Tennis Australia coaching brochure

Proposed collection

1. Record a clear, full-body clip

Keep the camera view consistent.

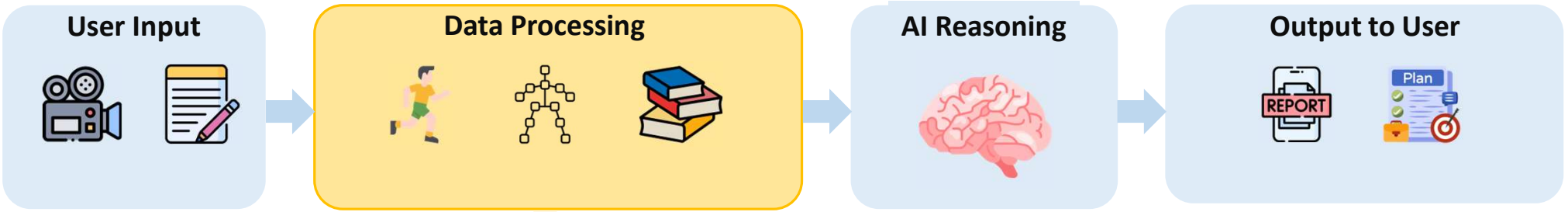
2. Attach player context

Stroke, level, goal and question

3. Prepare clips for processing

Label the stroke and key phases.
Check visibility and record the source.

Multimodal Data Understanding & Processing



*"Model performance is bounded by **data quality and representation**"*

Athlete Video



Pose Estimation



Pose Classification

Prediction: Forehand
Confidence: 90%

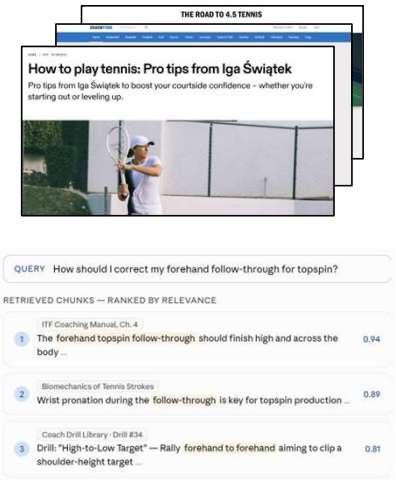
Wearable Sensors → **Tabular data**

Athlete #07 — Rohan S. | 10 Sep 2026 | Training session · 14:02–14:08 | IMU + BioSense v2

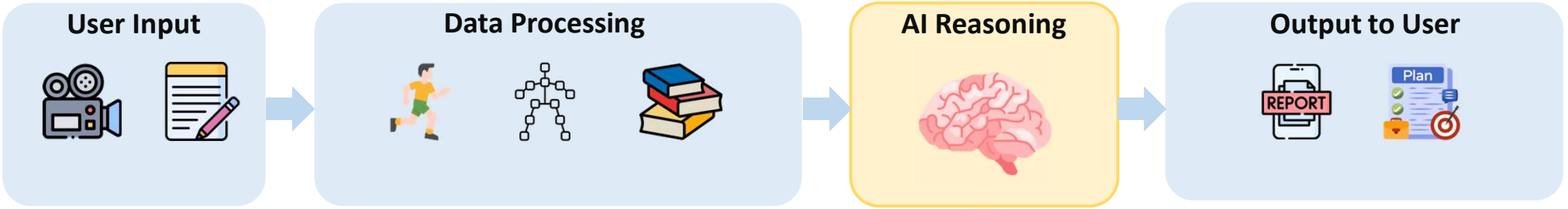
TIME	SENSOR	VALUE	UNIT	BODY SITE	STATUS
14:02:01	Heart rate	142	bpm	Wrist	Normal
14:02:01	SpO_2	98	%	Wrist	Normal
14:03:15	Accel. (X)	+2.31	m/s²	Lower back	Normal
14:03:15	Accel. (Z)	−8.74	m/s²	Lower back	Normal
14:04:30	Gyroscope	312	°/s	Right wrist	High
14:05:02	Skin temp.	37.8	°C	Forearm	Elevated
14:05:48	EMG	0.82	mV	Quad (R)	Normal
14:06:10	Heart rate	189	bpm	Wrist	⚠ Alert
14:07:22	Force plate	1,840	N	Right foot	High
14:08:05	GSR	14.3	µS	Palm	Normal

Knowledge Base

RAG ranked texts



AI Reasoning

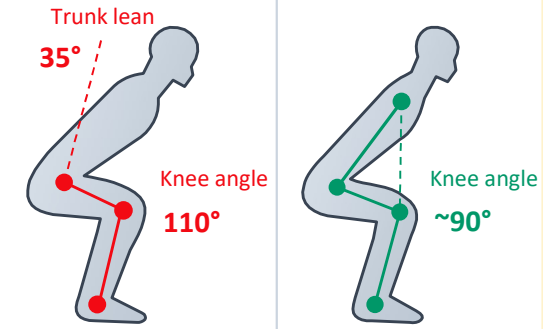


From measurements to meaningful coaching advice

CoT / Structured Reasoning Flow

① Read Input	Athlete performs squat; detected knee angle 110°, trunk lean 35°.
② Compare	Target knee angle about 90°, knees aligned with toes.
③ Diagnose	Squat is slightly shallow; knee moves inward.
④ Explain	May reduce stability and movement quality.
⑤ Recommend	Squat deeper gradually; keep knees aligned with toes.

Squat Example



Interactive Q&A

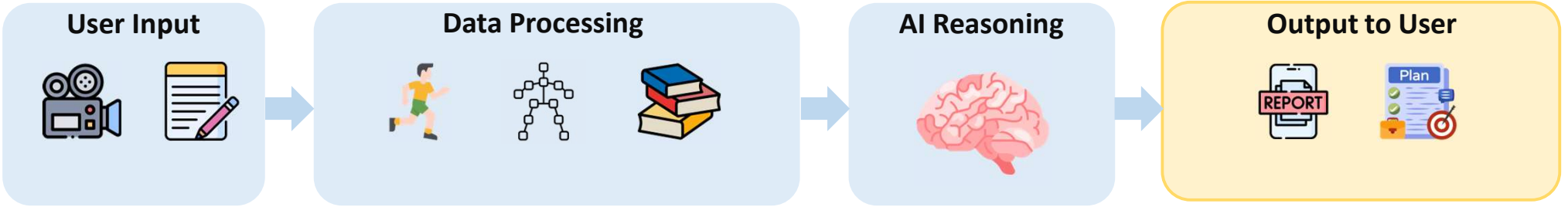
Why is knee alignment important?

It helps maintain more stable and safer squat movement.

What should I practice?

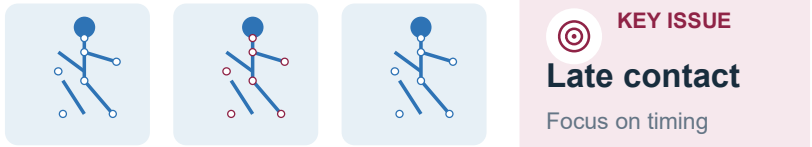
Try 3 x 10 slow squats and focus on knee-to-toe alignment.

Output to User



Performance feedback and a plan for the next practice

Stroke phase review

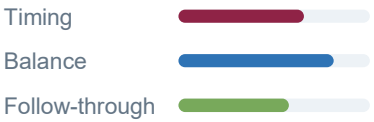


KEY ISSUE

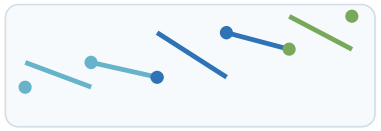
Late contact

Focus on timing

Technique profile



Workload and recovery



Safety note
Risk and recovery

Evidence link
Relevant coaching clip and source

WEEKLY OBJECTIVE

Earlier forehand contact

Practice schedule



Adaptive update



Review each session

Drill prescription

